

Homework Assignment 3

Solution Manual

Question 1 Reading a function from a relation matrix

(a) The relation as ordered pairs.

A matrix entry is 1 exactly when its row and column labels form an ordered pair in the relation. Therefore

$$R = \{(a, 2), (b, 1), (c, 2)\}.$$

(b) Function properties.

Each element of A occurs exactly once as a first coordinate. Equivalently, each row of the matrix contains exactly one 1. Thus R defines the function

$$f(a) = 2, \quad f(b) = 1, \quad f(c) = 2.$$

Its range is $\{1, 2\}$. It is not one-to-one because $a \neq c$ but $f(a) = f(c) = 2$. It is not onto B because neither 3 nor 4 is attained.

(c) Matrix conditions.

For a relation from X to Y :

- It is a function exactly when every row contains exactly one 1.
- A function is one-to-one exactly when every column contains at most one 1.
- A function is onto exactly when every column contains at least one 1.

Consequently, a function is bijective exactly when every row and every column contains exactly one 1.

Question 2 Composition, domains, and inverse behavior

(a) Computing $g \circ f$.

For every $x \in \mathbb{R}$,

$$(g \circ f)(x) = g(x^2 + 1) = \sqrt{(x^2 + 1) - 1} = \sqrt{x^2} = |x|.$$

The principal square root is nonnegative. Thus it equals x when $x \geq 0$ and $-x$ when $x < 0$; for example, $(g \circ f)(-2) = 2 \neq -2$.

(b) Computing $f \circ g$.

For every $t \in [1, \infty)$,

$$(f \circ g)(t) = f(\sqrt{t-1}) = (\sqrt{t-1})^2 + 1 = t.$$

Thus $f \circ g$ is the identity on $[1, \infty)$.

(c) Are the functions inverses?

No. Inverse functions require both composition identities. Although $f \circ g = \text{id}_{[1, \infty)}$, the other composition is $g \circ f = |\cdot|$, not $\text{id}_{\mathbb{R}}$. The function f is not one-to-one on $\mathbb{R}$, since $f(x) = f(-x)$.

(d) Restricting the domain.

Define

$$f_+ : [0, \infty) \rightarrow [1, \infty), \quad f_+(x) = x^2 + 1.$$

Then for $x \geq 0$,

$$(g \circ f_+)(x) = \sqrt{x^2} = x,$$

and for $t \geq 1$,

$$(f_+ \circ g)(t) = (\sqrt{t-1})^2 + 1 = t.$$

Hence $g = f_+^{-1}$.

Question 3 Proving properties of compositions**(a) If $g \circ f$ is one-to-one, then f is one-to-one.**

Suppose $f(x) = f(x')$ for $x, x' \in A$. Applying g to equal values gives

$$g(f(x)) = g(f(x')).$$

Since $g \circ f$ is one-to-one, $x = x'$. Therefore f is one-to-one.

(b) If $g \circ f$ is onto, then g is onto.

Take an arbitrary $z \in C$. Surjectivity of $g \circ f$ gives an $x \in A$ such that

$$g(f(x)) = z.$$

Let $y = f(x) \in B$. Then $g(y) = z$. Every $z \in C$ therefore has a preimage under g , so g is onto.

(c) Counterexamples to the converses.

One example disproves both converses. Let

$$A = \{a_1, a_2\}, \quad B = \{b_1, b_2, b_3\}, \quad C = \{c_1, c_2\},$$

and define

$$\begin{aligned} f &= \{(a_1, b_1), (a_2, b_2)\}, \\ g &= \{(b_1, c_1), (b_2, c_1), (b_3, c_2)\}. \end{aligned}$$

The function f is one-to-one because its two inputs have different outputs. The function g is onto because c_1 and c_2 are both attained. However,

$$g \circ f = \{(a_1, c_1), (a_2, c_1)\}.$$

This composition is not one-to-one because its two inputs have the same output, and it is not onto C because it misses c_2 .

Question 4 Reflexive, symmetric, and transitive relations

(a) Inclusion on $\mathcal{P}(U)$.

The relation is **reflexive**: every set $X \subseteq U$ satisfies $X \subseteq X$, so XX .

It is not symmetric. For example,

$$\emptyset \subseteq \{1\},$$

so $\emptyset R \{1\}$, but $\{1\} \not\subseteq \emptyset$, so $\{1\} R \emptyset$ is false.

It is **transitive**: if XRY and YRZ , then $X \subseteq Y$ and $Y \subseteq Z$. Every element of X is therefore an element of Y and hence of Z , so $X \subseteq Z$ and XRZ .

(b) Additive inverses on $\mathbb{Z}$.

The relation is not reflexive. For example, $1 + 1 \neq 0$, so $1S1$ is false.

It is **symmetric**: if xSy , then $x + y = 0$. Since addition is commutative, $y + x = 0$, so ySx .

It is not transitive. We have

$$1S(-1) \quad \text{and} \quad (-1)S1,$$

because each corresponding sum is 0. But $1S1$ is false because $1 + 1 = 2 \neq 0$.

(c) Equal-length binary strings.

The relation is **reflexive**: every string has the same length as itself.

It is **symmetric**: if u and v have the same length, then v and u have the same length.

It is **transitive**: if u and v have the same length and v and w have the same length, then

$$|u| = |v| = |w|,$$

so u and w have the same length. Thus uTw .

Question 5 An equivalence relation and its matrix blocks

(a) Proving the three properties.

For every $(a, b) \in X$, we have $a + b = a + b$, so $(a, b) \sim (a, b)$. The relation is reflexive.

If $(a, b) \sim (c, d)$, then $a + b = c + d$. Reversing the equality gives $c + d = a + b$, so $(c, d) \sim (a, b)$. The relation is symmetric.

If $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$, then

$$a + b = c + d \quad \text{and} \quad c + d = e + f.$$

Therefore $a + b = e + f$, so $(a, b) \sim (e, f)$. The relation is transitive. It is consequently an equivalence relation.

(b) The equivalence classes.

The classes are determined by the possible coordinate sums 0, 1, 2, 3, 4:

$$\begin{aligned} C_0 &= \{(0, 0)\}, \\ C_1 &= \{(0, 1), (1, 0)\}, \\ C_2 &= \{(0, 2), (1, 1), (2, 0)\}, \\ C_3 &= \{(1, 2), (2, 1)\}, \\ C_4 &= \{(2, 2)\}. \end{aligned}$$

These five sets are disjoint and together contain all nine elements of X .

(c) Ordering by equivalence class.

One valid ordering is

$$(0, 0); \quad (0, 1), (1, 0); \quad (0, 2), (1, 1), (2, 0); \quad (1, 2), (2, 1); \quad (2, 2).$$

The order within each semicolon-delimited class may be changed.

(d) The compressed matrix.

In the displayed ordering, the matrix is

$$M_{\sim} = \text{diag}(J_1, J_2, J_3, J_2, J_1) = \begin{pmatrix} J_1 & 0 & 0 & 0 & 0 \\ 0 & J_2 & 0 & 0 & 0 \\ 0 & 0 & J_3 & 0 & 0 \\ 0 & 0 & 0 & J_2 & 0 \\ 0 & 0 & 0 & 0 & J_1 \end{pmatrix}.$$

The block sizes 1, 2, 3, 2, 1 are the sizes of the five equivalence classes. Every two elements inside one class have the same coordinate sum, so each diagonal block contains only 1s. Elements in different classes have different sums and are not related, so every off-diagonal block is zero.

Question 6 Countability of finite Cartesian powers of $\mathbb{N}$ **(a) Encoding and decoding a triple.**

First,

$$H(0, 1) = 2^0(2 \cdot 1 + 1) - 1 = 2.$$

Therefore

$$K_3(0, 1, 2) = H(2, 2) = 2^2(2 \cdot 2 + 1) - 1 = 19.$$

To decode 19, add 1 and factor the result into a power of 2 times an odd number:

$$19 + 1 = 20 = 2^2 \cdot 5.$$

Thus the first input to the outer copy of H was 2, and its second input was $(5 - 1)/2 = 2$. We have recovered

$$H(a, b) = 2, \quad c = 2.$$

Decode 2 in the same way:

$$2 + 1 = 3 = 2^0 \cdot 3.$$

Hence $a = 0$ and $b = (3 - 1)/2 = 1$. The recovered triple is $(0, 1, 2)$.

(b) Proving K_3 is a bijection.

One-to-one. Suppose

$$K_3(a, b, c) = K_3(a', b', c').$$

Then

$$H(H(a, b), c) = H(H(a', b'), c').$$

Injectivity of H gives

$$H(a, b) = H(a', b') \quad \text{and} \quad c = c'.$$

Applying injectivity of H again gives $a = a'$ and $b = b'$. Thus the triples are equal, so K_3 is one-to-one.

Onto. Take any $n \in \mathbb{N}$. Since H is onto, there are $p, c \in \mathbb{N}$ such that

$$H(p, c) = n.$$

Since H is onto again, there are $a, b \in \mathbb{N}$ such that $H(a, b) = p$. Then

$$K_3(a, b, c) = H(H(a, b), c) = H(p, c) = n.$$

Therefore K_3 is onto. It is a bijection, so $\mathbb{N}^3$ is countably infinite.

(c) Generalization to $\mathbb{N}^k$.

Define $K_1 : \mathbb{N} \rightarrow \mathbb{N}$ by $K_1(x_1) = x_1$. Recursively, once $K_k : \mathbb{N}^k \rightarrow \mathbb{N}$ has been defined, set

$$K_{k+1}(x_1, \dots, x_k, x_{k+1}) = H(K_k(x_1, \dots, x_k), x_{k+1}).$$

We prove by induction on $k \geq 1$ that K_k is a bijection.

Base case ($k = 1$). The function K_1 is the identity on $\mathbb{N}$, so it is a bijection.

Inductive step. Assume K_k is a bijection.

To prove injectivity of K_{k+1} , suppose

$$K_{k+1}(x_1, \dots, x_{k+1}) = K_{k+1}(y_1, \dots, y_{k+1}).$$

Injectivity of H gives

$$K_k(x_1, \dots, x_k) = K_k(y_1, \dots, y_k) \quad \text{and} \quad x_{k+1} = y_{k+1}.$$

Injectivity of K_k then gives $x_i = y_i$ for $1 \leq i \leq k$. Thus the two $(k + 1)$ -tuples are equal.

To prove surjectivity, take $n \in \mathbb{N}$. Since H is onto, choose $p, q \in \mathbb{N}$ with $H(p, q) = n$. Since K_k is onto, choose $(x_1, \dots, x_k) \in \mathbb{N}^k$ with $K_k(x_1, \dots, x_k) = p$. Then

$$K_{k+1}(x_1, \dots, x_k, q) = H(p, q) = n.$$

So K_{k+1} is also a bijection. Induction proves that $\mathbb{N}^k$ is countably infinite for every integer $k \geq 1$.

The case $k = 0$ is excluded under the convention $\mathbb{N} = \{0, 1, 2, \dots\}$: the set $\mathbb{N}^0$ contains only the empty tuple, so it is finite rather than countably infinite.